

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sep 2021 – Apr 2026

TECHNICAL SKILLS

Languages: Python, C/C++

Systems: Linux, Bash, Git, Windows, ROCm, Docker

Validation: RAS, IO, Performance, Memory, System Management

Integration: BMC, BIOS, firmware, system integration

Data: SQL, IBM DB2, NumPy, pandas

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, BMC, BIOS, Linux, Bash, Git, Windows, ROCm, Docker, RAS, IO, Performance, Memory, System Management, firmware, system integration

Markham, ON

Datacenter GPU System Integration Engineer (Co-op)

May 2024 – Aug 2025

- Ran automated post-silicon diagnostic validation across MI300X, MI325X, and MI350X **Linux** and **Windows** platforms against validated BKC's, sustaining **Performance, IO, RAS, Memory, and System Management** coverage while exercising datacenter GPU **system integration** paths at roughly 100+ executions weekly.
- Deployed and maintained **Linux** and **Windows** GPU lab stacks spanning **BMC, BIOS**, and OS images plus **Docker**-based workload environments for repeatable **system integration** bring-up, stressing **RAS, IO, Performance, Memory, and System Management**, trimming early bring-up failures by about 20%.
- Extended **Python** test automation through an internal workload launcher and **Git**-tracked utilities so engineers reproduced diagnostics consistently, dropping recurring manual setup from about 20 minutes to under 5 minutes.
- Triaged hardware and software defects spanning **ROCm, C/C++** firmware and driver stacks, and interconnect paths via systematic log review and scripted reproduction flows, cutting median time-to-root-cause on assigned defects by roughly 30%.
- Worked with infrastructure, firmware, and validation teams to track work in Jira and improve checklists and documentation, which made complex GPU diagnostic tests easier to reproduce and reduced misconfigurations in new test setups.

PROJECTS

Linux Log Normalization and Token Frequency Analysis | Python, NumPy, pandas

January 2026

- Developed a configurable text normalization pipeline in **Python** for processing raw **Linux logs**, supporting tokenization, stemming, lemmatization, and digit filtering.
- Performed token frequency computation and statistical aggregation using **NumPy** and **pandas** to analyze distributional patterns in **Linux logs**.

SufSort Sorting System | C, Bash, Linux

November 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Ontario Hospital Association Database Design | SQL, IBM DB2

September 2025

- Authored SQL scripts implementing primary keys, foreign keys, and referential integrity constraints to ensure consistency across patient and record data.
- Developed a full database creation script compatible with **IBM DB2** CAS server environments for reproducible deployment.

RELEVANT COURSES

Software Testing, Computer Performance Analysis, Real-Time Systems, Computer Architecture, Database Systems